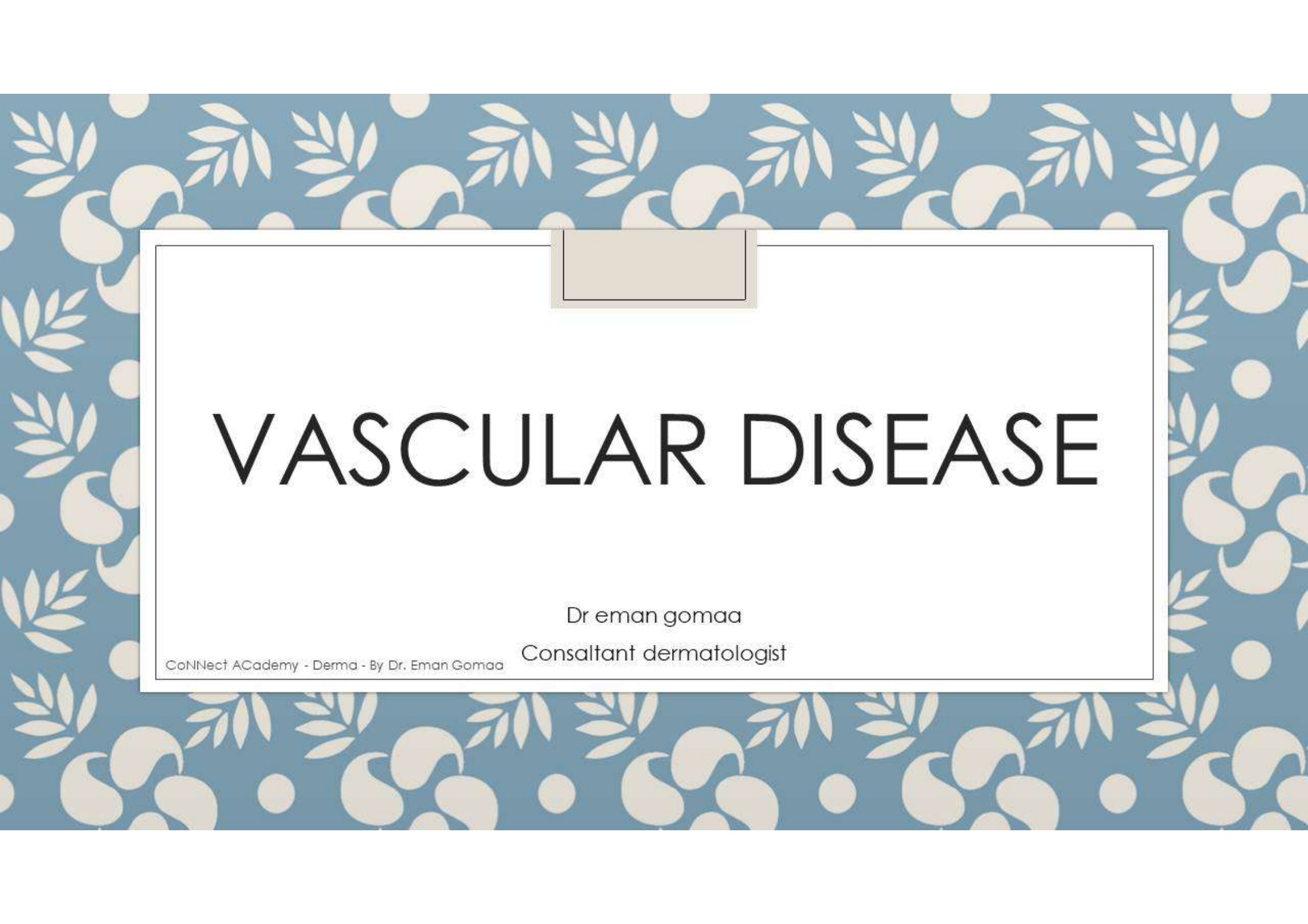


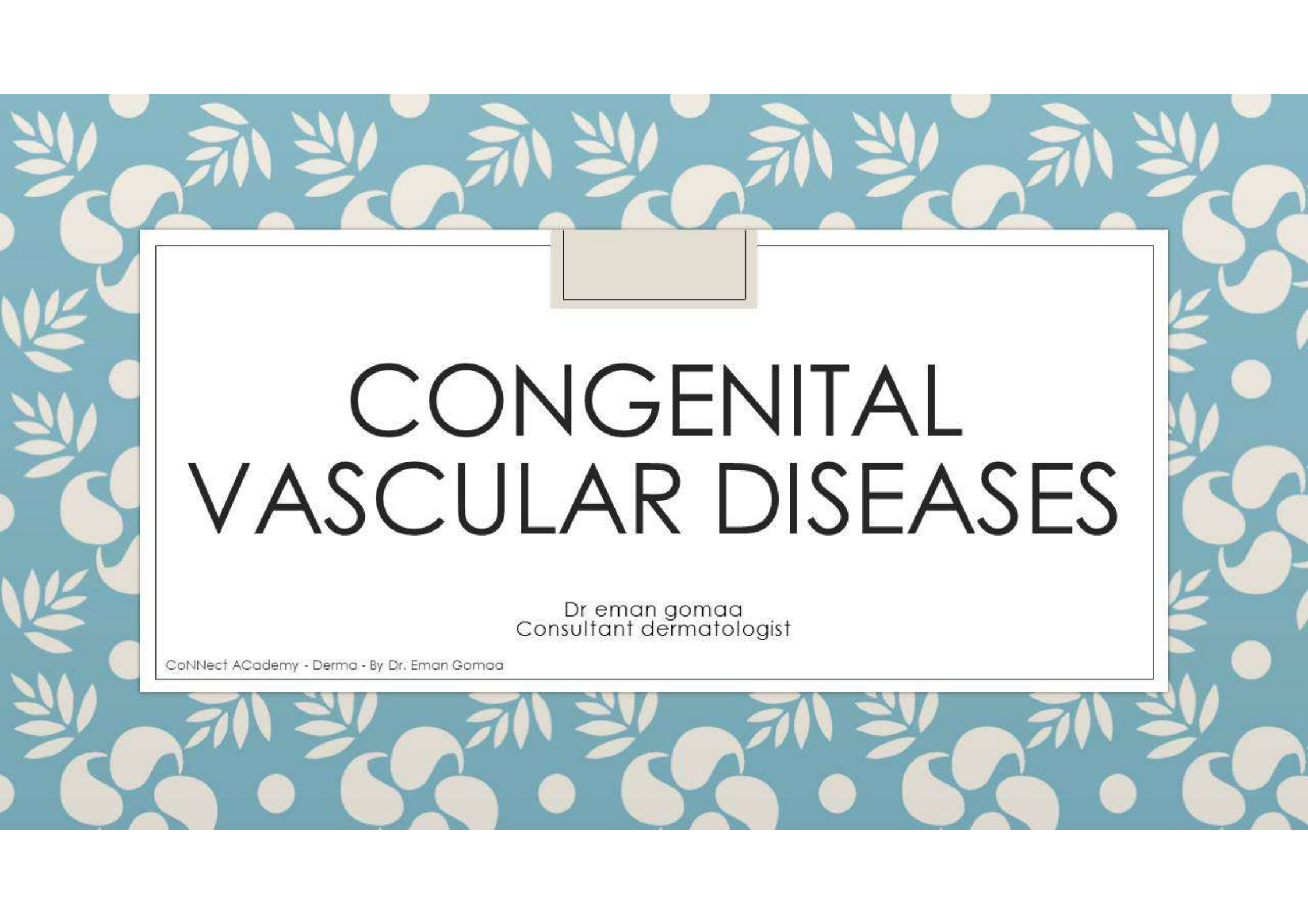


VASCULAR DISEASE

Dr eman gomaa

Consaltant dermatologist

CoNNect ACademy - Derma - By Dr. Eman Gomaa



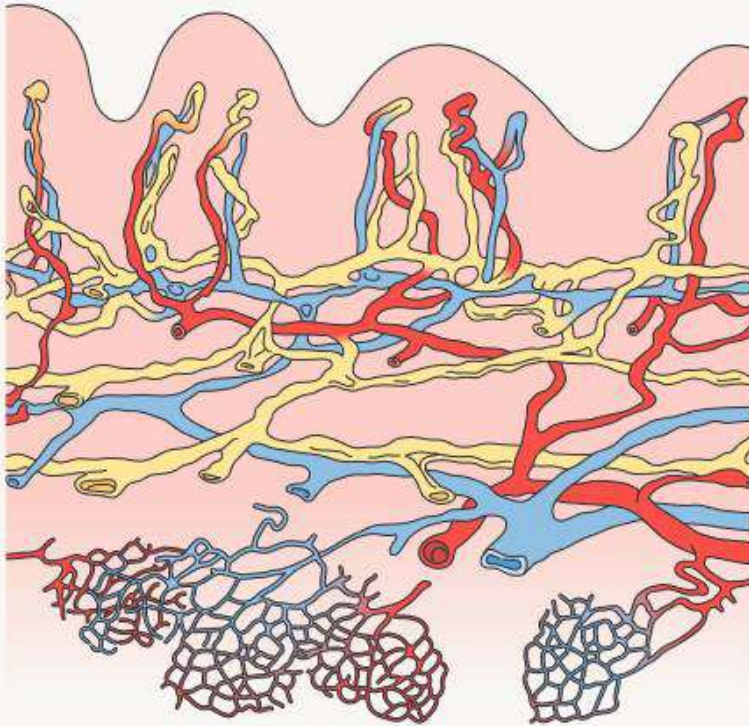
CONGENITAL VASCULAR DISEASES

Dr eman goma
Consultant dermatologist

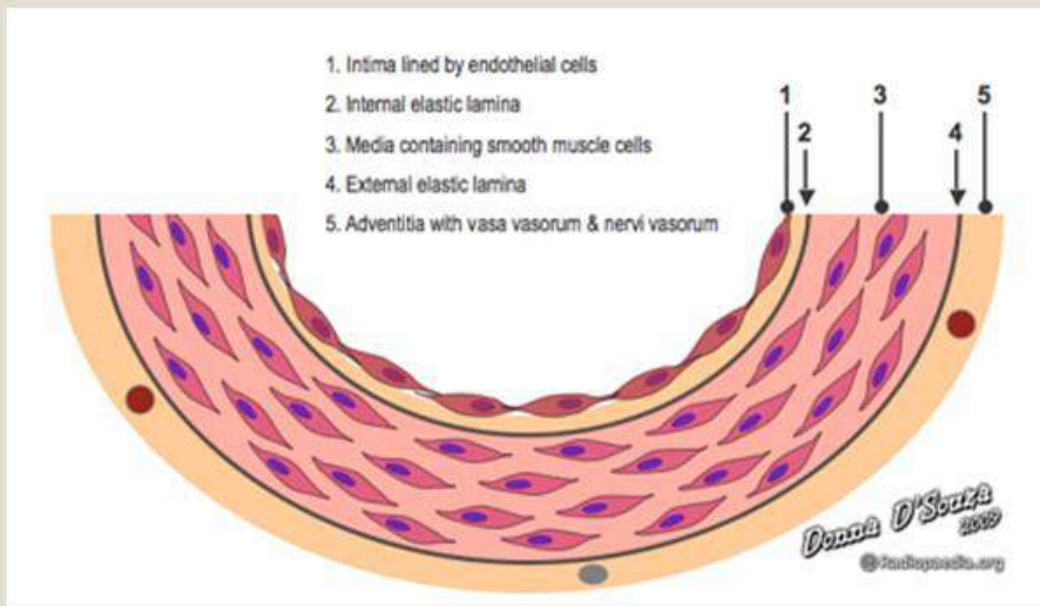
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The dermal microvascular unit

SCHEMATIC REPRESENTATION OF THE CUTANEOUS VASCULAR SYSTEM

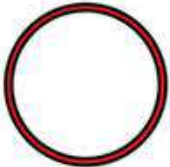


- The dermal microvasculature is divided into 2 plexuses
-
- (1) The superficial vascular plexus:
 - Located at the boundary between the papillary and the reticular dermis.
 - Composed of anastomosing arterioles and venules.
- (2) The deep vascular plexus:
 - A Located at the boundary between the reticular dermis and the subcutaneous fat.



- **The wall of the small arteries and the arterioles have 3 layers:**
 1. The intima: composed of endothelial cells and an internal elastic lamina.
 2. The media: composed of smooth muscle cells.
 3. The adventitia: composed of connective tissue.

Classification of blood vessels according

	AORTA	ARTERY	ARTERIOLE	CAPILLARY	VENULE	VEIN
						
Human	25,000/2,000	4,000/1,000	30/20	8/1	20/2	5,000/500
Mouse	535/55	150/50	18/4.7	4/0.3	14/1	250/50

○

1. small vessels:

- Include arterioles, capillaries and post-capillary venules.

© Located in the superficial and mid-dermis of the skin.

○

2. Medium -sized vessels:

- ® Include small arteries and veins.

© Located in the deep dermis or subcutis.

○

3. Large vessels:

- © Include the aorta and named arteries.

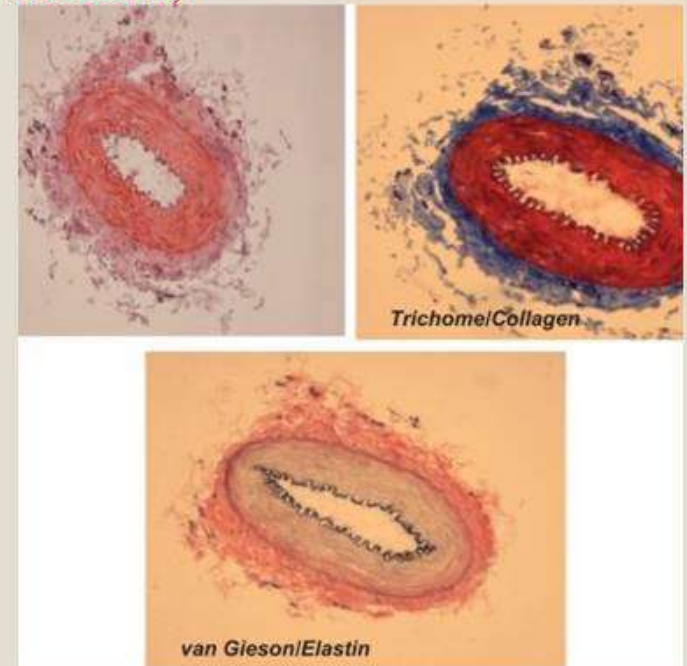
• NOT FOUND IN THE SKIN

◦ **Special stains:**

- Elastic fibers : are visualized **with van Gieson stain (black) or orcein stain (dark brown)**
- **Periodic acid-Schiff (PAS)** stain visualize the basement membrane

◦ **Immunohistochemistry :**

- **C D 31**: High specificity and good sensitivity for endothelial tumors.
- **C D 34**: High sensitivity but low specificity for endothelial tumors (positive also in dermatofibrosarcoma protuberant and some fibrous tumors).
- **Factor V I I I**-related antigen (von Willebrand factor; vW F

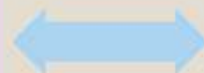


Vascular
birth mark



Tumor

malformation



Hemangioma

Capillary
Venouse
Lymphatic



DIFFERENCES BETWEEN INFANTILE HEMANGIOMAS AND VASCULAR MALFORMATIONS		
	Infantile hemangioma	Vascular malformation
Clinical	• Usually absent at birth or subtle precursor lesion present; only occasionally well formed • Rapid proliferation • Spontaneous involution over years	• Usually evident at birth • Slow expansion, proportionate with growth • Persists into adulthood
Epidemiology	More common in: • Girls (2–5:1) • Premature and/or low-birth-weight infants • Infants whose mothers were older or who underwent chorionic villus sampling	No gender or gestation predilection
Pathology	Proliferating: endothelial cell hyperplasia, lobule formation, mast cells, prominent basement membrane Involuting: fibrofatty tissue replacement, decreased mast cells when fully involuted	Dependent upon type, often irregular vascular channels
Immunohistochemistry	Positive for GLUT1, Lewis Y antigen, merosin, and FcγRII, Wilms tumor protein 1 (WT1)	Negative for GLUT1, Lewis Y antigen, merosin, and FcγRII, WT1

BIOLOGIC CLASSIFICATION OF VASCULAR BIRTHMARKS

Vascular tumors

Benign

- Infantile hemangioma
- Congenital hemangioma (rapidly involuting, non-involuting, partially involuting)
- Tufted angioma
- Pyogenic granuloma
- Spindle cell hemangioma
- Others

Locally aggressive or borderline

- Kaposiform hemangioendothelioma
- Others, e.g. retiform or composite hemangioendothelioma

Vascular malformations

Simple

- Capillary malformations (CMs)
- Venous malformations (VMs)
 - Common, glomuvenous, others
- Lymphatic malformations (LMs)
 - Common: macrocystic, microcystic, mixed; others
- Arteriovenous malformations, arteriovenous fistula (AVMs, AVFs; high flow)

Combined

- CVM, CLM, LVM, CAVM, CLAVM, others

Of major named vessels

Associated with other anomalies

- Sturge-Weber syndrome, Klippel-Trenaunay syndrome, others

Provisionally unclassified

- Angiokeratomas
- Multifocal lymphangioendotheliomatosis with thrombocytopenia
- Kaposiform lymphangiomatosis
- Others

infantile Hemangiomas



- infantile hemangiomas are benign tumors of endothelial cells. They are the most common tumors arising in the neonatal period and infancy.
- 4-5% of infant
- 1st few weeks of life
- Femal :male 9:1
- Risk factor :
 - Female
 - Premature
 - low birth
 - Old mother
 - Multigestation
 - Prenatal instrument



- **Age** : usually during the first few weeks of life . **absent at birth**
- **Site**: 50% of cases are located on the head and neck
- **c/p** : appearance determined by location :
 - **Superficial** 50-60%
 - **Deep** 15%
 - **Mixed** : 25 – 35 %





➤ **Superficial : 50- 60 %**

➤ bright red . Finely lobulated surface

erythematous patch >>>> papule >>> plaque





➤ **deep : 15 %**

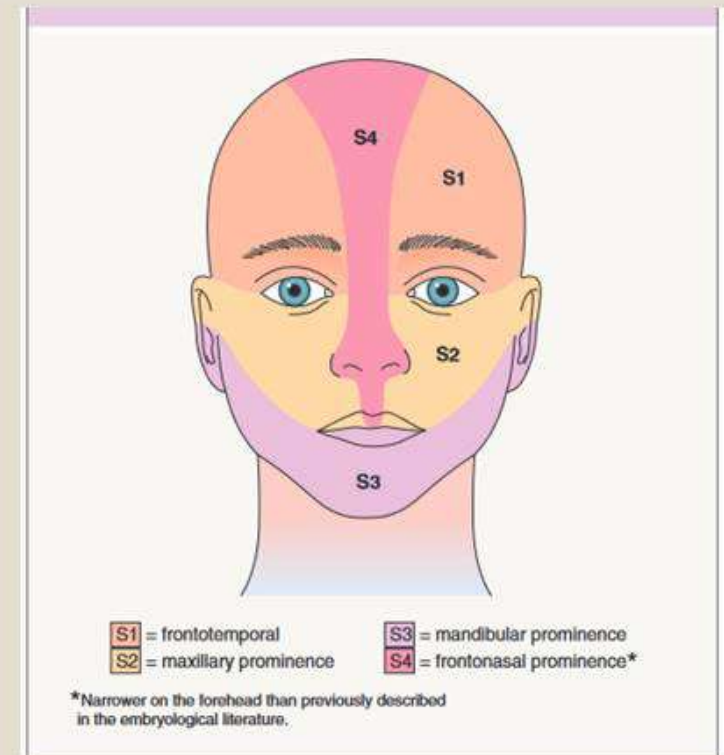
- **Deep dermis , sc ,**
- **Warm , ill defined light blue to purple nodules**
- **No skin changes**

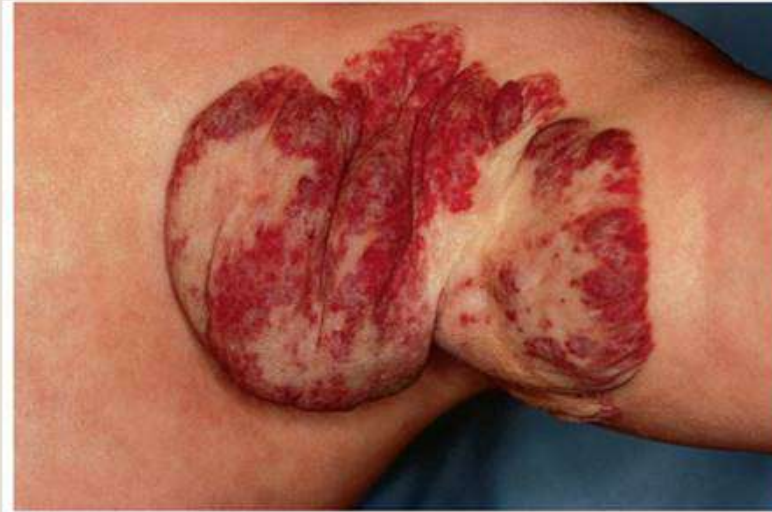


- **Mixed :**
 - Well defined vascular plaque
 - overlying larger poorly circumscribed light blue nodules

➤ Pattern of involvement

- Focal lesions (most commonly).
- Segmental lesions.
- Indeterminate lesions
- Most cases of infantile hemangiomas are solitary
- . 25% of patients have multiple lesions, which are occasionally associated with visceral hemangiomatosis.





- (Course) :
 - (1) The proliferative phase: increase in the size of the hemangioma.
 - (2) The involution phase :
 - begin at first year and continues for several years.
 - A color change from bright red to gray-purple and flattening of the surface are the earliest signs of involution



Fig. 103.10 Residua of infantile hemangiomas. Hemangioma with ulcer (A), which involuted leaving minimal hypopigmentation (arrows) and a circular scar at site of ulceration in the same patient 20 years later (B); C, D telangiectasias; and E atrophy and fibrofatty changes. A, B, Courtesy, Ronald P. Rapini, MD; C, D, Courtesy, Jean L. Bolognia, MD.

Histeopathology

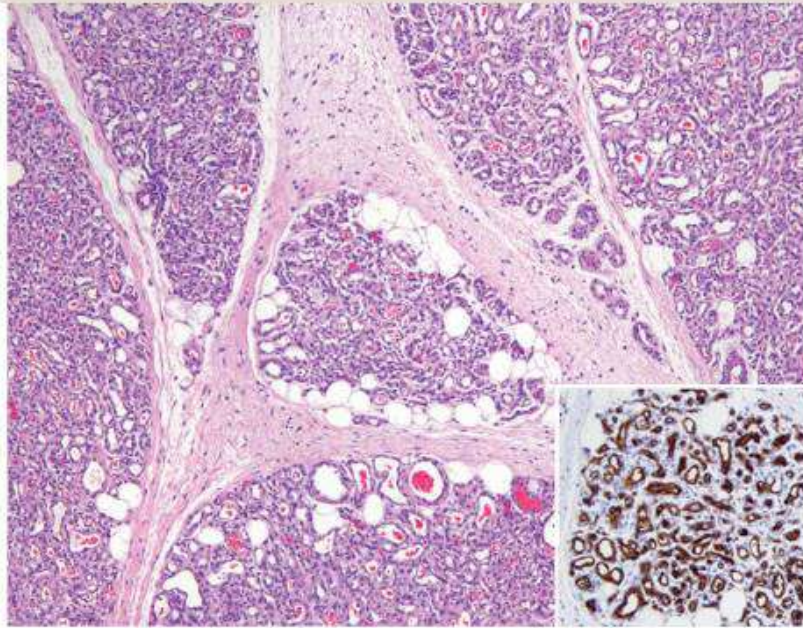


Fig. 103.21 Histologic features of a proliferating infantile hemangioma. Collections of endothelial cells forming vascular lumens, separated by fibrotic septae. The inset shows positive GLUT1 staining of endothelial cells. *Courtesy, Matthew Kuhar, MD.*

- **Proliferative phase :**
 - well-defined non-encapsulated **lobules** separated by **fibrous septae** and formed of masses of endothelial **cells and pericytes**. Small vascular lumens
- **Involution stag :**
 - flattening of the endothelial cells, wider lumens and fibrofatty tissue
 - Immunohistochemistry :
 - glucose transporter protein-1 (**GLUT-1**)

Complications of IH

- - **Ulceration**
- **Interference with function**
- **Associated comorbidities and syndromes**

- Ulcerating IH >>>>>>>>Bleeding , infection &scarring



Fig. 103.11 Ulcerated hemangiomas. **A** Ulcerated early superficial hemangioma on the buttock. Because the hemangioma component may not be obvious, this diagnosis should be considered when an infant presents with an ulcer in the diaper area. **B** Ulcerated thick superficial hemangioma above the ear. Note the distortion of the pinna. *Courtesy, Julie V Schaffer, MD.*

Interference with vital functions

- Periorbital IH >>>>>>> Amblyopia



- Nasal hemangioma >>>>>>> Difficult breathing, ulceration & deformity



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Lip Hemangioma>> Difficult feeding,ulceration



Hemangioma of the ear>>>> Conductive deafness



Perineal hemangioma >>> ulceration,painful defecation &urination

SYSTEMIC COMPLICATION

Internal organs Hs (specially hepatic)

- Upper facial H. → **PHACES syndrome**
- Lower facial H. → Laryngeal H.& Cardiac anomalies.
- Lumbar H. → **LUMBAR syndrome**
- Hemangioma-Haemorrhage syndrome (**Kasabach-Merritt syndrome**)



Fig. 103.19 Multifocal infantile hemangiomas with extracutaneous hemangiomas. In this child, many small, superficial skin lesions were present in a miliary pattern and hemangiomas were present in the liver. Historically, this was referred to as "diffuse neonatal hemangiomatosis". *Courtesy, Julie V Schaffer, MD.*



Fig. 103.20 Multifocal hepatic hemangiomas. CT scan demonstrating multiple hepatic hemangiomas in a child with multifocal cutaneous infantile hemangiomas ("diffuse neonatal hemangiomatosis"). *Courtesy, Arny Palter, MD.*

Large or disseminated cutaneous Hs lesions 10(>

Visceral Hs (Specially hepatic)

- HF (HCOP failure): 30-80%
- Hypothyroidism

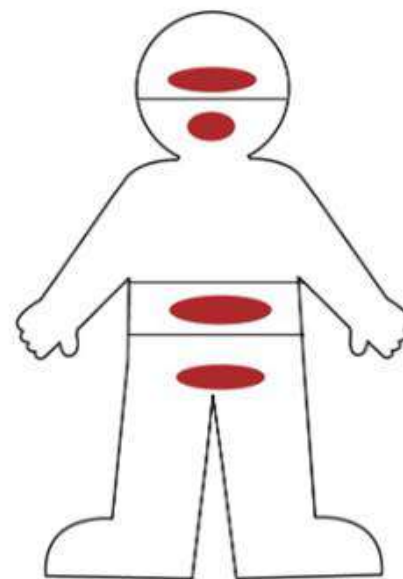


.Upper Facial H
PHACES syndrome

.Lower facial H
Laryngeal & Cardiac Hs

.Lumbar H
LUMBAR syndrome

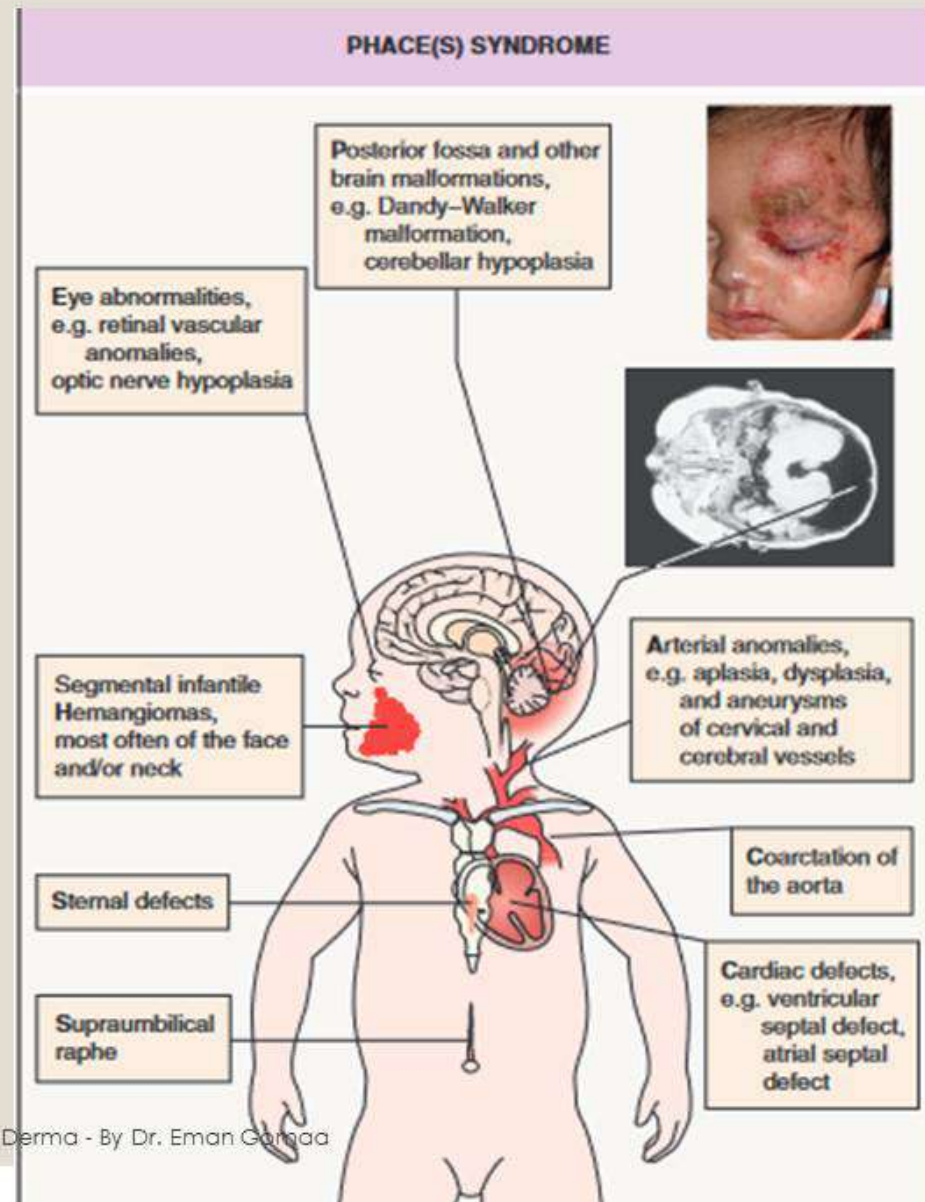
.Pelvic H
PELVIS syndrome



Phaces syndrom



Fig. 103.12 Segmental hemangioma with significant periorbital component partially obstructing the visual axis. Courtesy, Julie V Schaffer, MD.





DIAGNOSTIC CRITERIA FOR PHACE(S) SYNDROME

Definite PHACE(S):

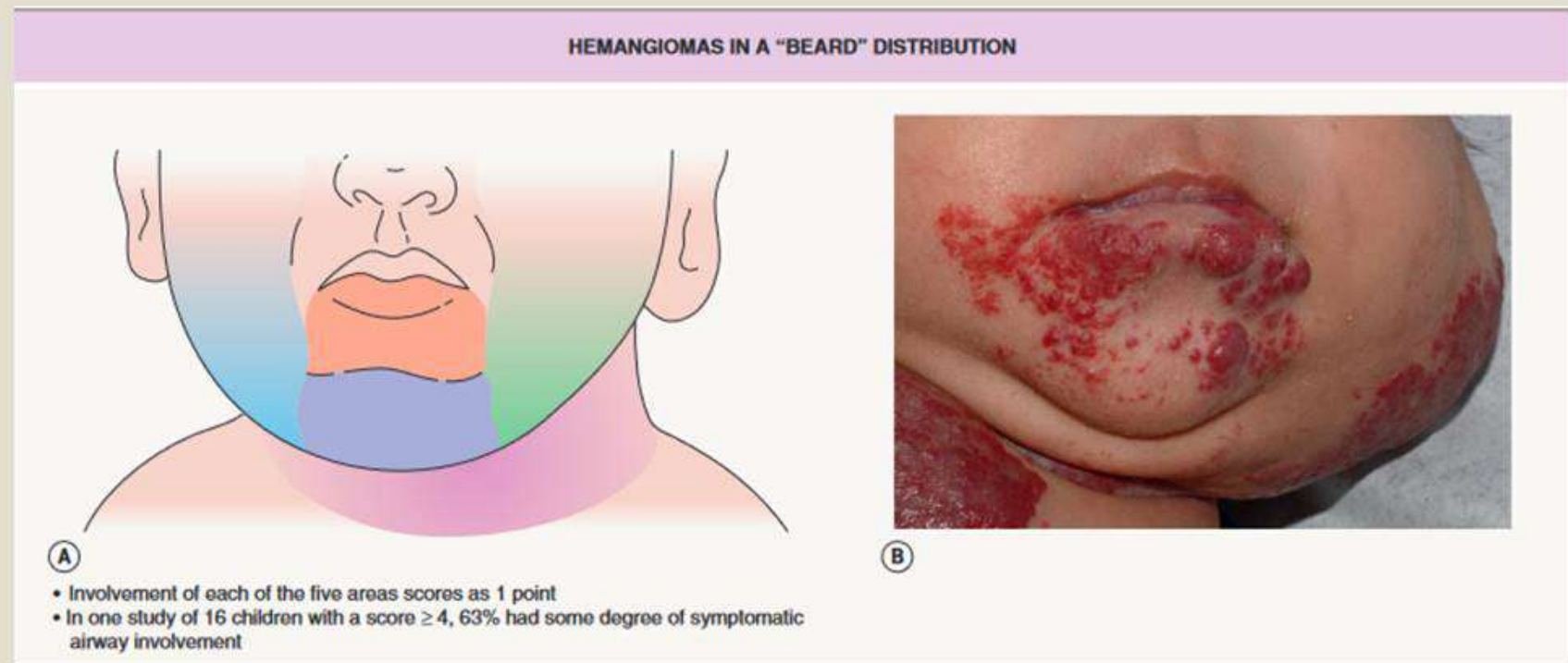
- Hemangioma >5 cm in diameter of the head including scalp PLUS 1 major criterion or 2 minor criteria
- Hemangioma of the neck, upper trunk, or trunk and proximal upper extremity PLUS 2 major criteria

Possible PHACE(S):

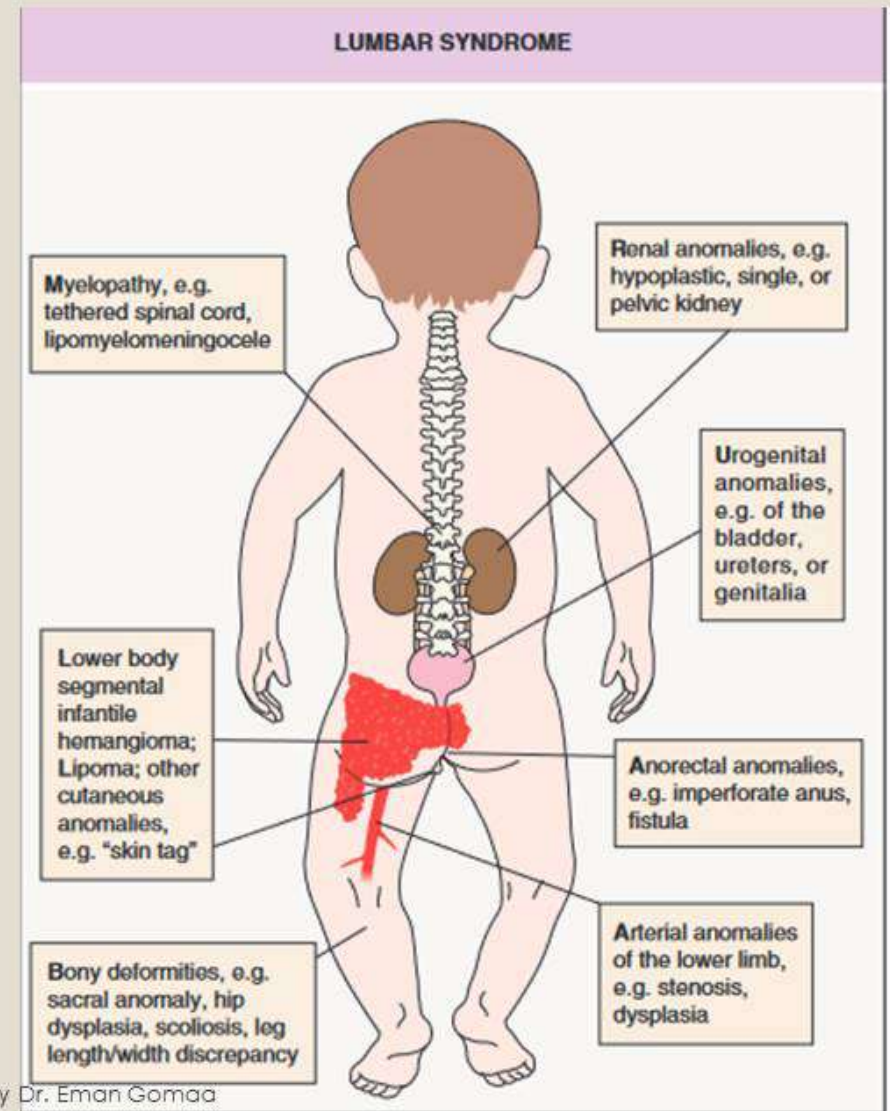
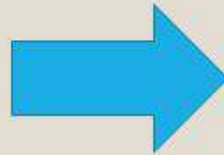
- Hemangioma >5 cm in diameter of the head including scalp PLUS 1 minor criterion
- Hemangioma of the neck, upper trunk, or trunk and proximal upper extremity PLUS 1 major criterion or 2 minor criteria
- No hemangioma PLUS 2 major criteria

Organ system	Major criteria	Minor criteria
Arterial	• Anomalies of major cerebral or cervical arteries* – dysplasia**, hypoplasia, stenosis/occlusion, aberrant origin/course • Persistent carotid-vertebrobasilar anastomosis (e.g. proatlantal segmental, hypoglossal, otic, trigeminal arteries)	• Aneurysm of cerebral arteries
Structural brain	• Posterior fossa anomalies – Dandy-Walker complex, other hypoplasia/dysplasia of the mid or hind brain	• Midline anomalies • Malformation of cortical development
Cardiovascular	• Aortic arch anomalies • Aberrant origin of subclavian artery ± vascular ring	• Ventricular septal defect • Right/double aortic arch • Systemic venous anomalies
Ocular	• Posterior segment anomalies – persistent hyperplastic primary vitreous/ persistent fetal vasculature, retinal vascular anomalies, optic nerve hypoplasia, morning glory disc anomaly	• Anterior segment anomalies, e.g. sclerocornea, cataract, coloboma, microphthalmia
Ventral or midline	• Sternal defect/pit/cleft • Supraumbilical raphe	• Hypopituitarism • Ectopic thyroid • Midline sternal papule/hamartoma

Lower facial H. → Laryngeal H.& Cardiac anomalies, laryngeal obstruction



Lumbar H. → **LUMBAR** syndrome



(Kasabach-Merritt syndrome)

- is the association of a vascular tumor with a thrombocytopenic coagulopathy
- . It is a life-threatening complication
- **rapidly enlarging, ecchymotic patch with thrombocytopenia and a consumptive coagulopathy**



Fig. 103.22 Kaposiform hemangioendothelioma associated with Kasabach-Merritt phenomenon in a 5-week-old boy. Progressive induration and ecchymosis developed rapidly over a 24-hour period. An eczematous eruption is also evident. *Courtesy, Julie V Schaffer, MD.*

DD

- Hemangioma precursor >>>> port-wine stain “PWS”
- Deep hemangiomas >> vascular malformation (venous, lymphatic or combined)
- Superficial hemangiomas >>>> pyogenic granulomas or Spitz’s nevus

Treatment :

- choice of a particular line of treatment is dependent on many factors including:
 - *Size*
 - *Location*
 - *Interference with function*
 - *Presence or absence of systemic affection*
 - *Presence or absence of ulceration*

Treatment of IH

, Small, Localized
uncomplicated

➤ **Reassurance** : (About
natural course & spontaneous
resolution).

➤ **TOPICALS:**

- Beta-Blockers (Timolol
EDs).
- Corticosteroids (Topical
& intralesional).
- Imiquimod

Large, Complicated

1st Line

➤ **SYSTEMIC:**

- Propranolol
- Corticosteroids

2nd Line

- IFNα2b
- LASER
- Surgery
- Others

- ***1) NO treatment (Non-intervention):***

- Explanation to parents, periodic observation and follow-up

- .

- (2) Topical treatment:***

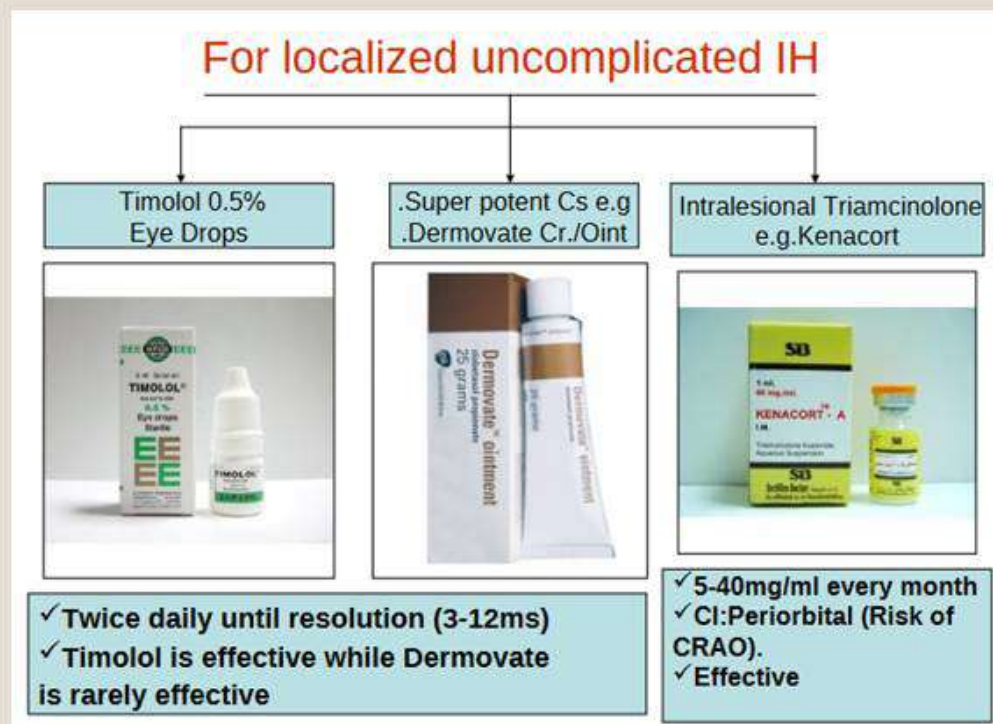
- Topical corticosteroids: super-potent
- Topical p-blockers: timolol (non-selective P-blocker) 0.5% gel or solution.
- Topical imiquimod 5%

(3

◦ 3) **Intralesional corticosteroids:**

- Indications: small localized lesions (to avoid systemic absorption).

Drug and dose: triamcinolone acetonide 5 - 40 mg/ml (the total maximum dose should not exceed 3-5 mg/kg per treatment session)



4) Systemic therapy

REASONS TO CONSIDER SYSTEMIC THERAPY FOR INFANTILE HEMANGIOMAS

Threatened vital functions

- Vision
- Airway

Potential for disfigurement

- Nasal tip/columella
- Lip, especially if the lesion crosses the vermilion border
- Large/rapidly growing lesion, especially if on the face

Severe/recalcitrant ulceration

High-output cardiac failure

- **Systemic corticosteroids:**

- : prednisolone 2-3 mg/kg/day to 3-5 mg/kg/day.
- corticosteroids suppress (VEGF) production
- inhibit vasculogenesis
- .

○ Systemic B-blockers:

- propranolol (non-selective B-blocker) 2 mg/kg/day divided into 2 or 3 oral doses.
 - vasoconstriction,
 - decreased expression of pro-angiogenic factors (e.g. VEGF, bFGF),
 - induction of endothelial cellapoptosiS..
 - • Possible adverse reactions: hypotension, bradycardia, hypoglycemia, bronchospasm and sleep disturbances.
 - o A baseline cardiac assessmen is essential prior to propranolol therapy.
- Vincristine:

.Systemic Propranolol for Hs



ADMINISTRATION OF PROPRANOLOL FOR INFANTILE HEMANGIOMAS (IH): PRETREATMENT EVALUATION, DOSING, AND MONITORING

Pretreatment evaluation

- *If history of bronchospasm/reactive airway disease* (a relative contraindication to propranolol therapy): consult with a pulmonologist or consider alternate therapy
- *If at risk for PHACE(S) syndrome**: MRI/MRA of head/neck and echocardiogram to exclude arterial and cardiovascular anomalies (see Fig. 103.15); if abnormal, consult with a neuroradiologist, neurologist, and/or cardiologist as indicated
- Cardiac examination, heart rate, and manual blood pressure; also consider an electrocardiogram (ECG); bradycardia (<80 beats per minute), greater than first degree heart block, blood pressure $<50/30$ mm Hg, and decompensated heart failure are considered to be contraindications

Dosing and course

- Initial: usually ~ 1 mg/kg/day divided into 2 (or 3) doses**
- Goal: 2–3 mg/kg/day divided into 2 (or 3) doses**; the target dose for the FDA-approved formulation (Hemangeol™) is as high as 3.4 mg/kg/day
- For a proliferating IH, treatment is typically continued for 6–12 months, depending on the clinical setting and course†
 - The dose should be adjusted to account for weight gain and maintain the goal dose/kg, especially during the proliferative phase
- Taper over 2–4+ weeks before discontinuing, primarily to assess for rebound growth

Monitoring

- Risks of propranolol include bradycardia, hypotension, hypoglycemia, sleep disturbances, and bronchospasm
- Consider a brief inpatient stay (e.g. 1–2 days) with telemetry for initiation of treatment if <5 weeks of age (using corrected age for premature infants) and/or complications such as cardiovascular or cerebrovascular anomalies
- Outpatient initiation of therapy
 - Monitor heart rate and blood pressure for 2 hours after the initial dose or a significant dose increase
 - Measure heart rate and blood pressure every 4–8 weeks while on a stable dose
 - Normal ranges for pediatric vital signs are available at: www.emedicinehealth.com/pediatric_vital_signs/article_em.htm
- Educate parents about signs of hypoglycemia – early (may be masked by β -blockade): sweating, rapid heart rate, shakiness, fussiness; late: lethargy, poor feeding, hypothermia, seizures
- Administer during or immediately after a feeding; twice daily doses should be ≥ 9 hours apart
 - Dose should be lowered or therapy suspended during intercurrent illnesses with decreased oral intake or vomiting

- ***Vincristine***

- • A chemotherapeutic agent that interferes with microtubule formation during mitosis, inducing apoptosis of tumor and endothelial cells.

- **(5) Laser therapy:**

- Pulsed dye laser
- The Nd:YAG

- **6) surgery**

N.B

KEY FEATURES OF INFANTILE AND CONGENITAL HEMANGIOMAS		
Infantile hemangioma	Rapidly involuting congenital hemangioma (RICH)	Non-involuting congenital hemangioma (NICH)
Usually absent or precursor lesion at birth	Fully developed at birth	Fully developed at birth
Rapid postnatal proliferation	Intrauterine proliferation	Intrauterine proliferation then proportionate postnatal growth
Slow spontaneous involution	Rapid involution during the first year*	Do not involute spontaneously
More common in girls	Equal prevalence or slightly more common in girls	Equal prevalence or slightly more common in girls
Lobular endothelial proliferation during proliferative phase; fibrous and fatty tissue during involution	Capillary lobules within fibrotic stroma containing thin-walled vessels, hemosiderin	Lobules of small, thin-walled vessels with large central vessel; dilated, dysplastic veins between lobules; "hobnailed" endothelial cells lining intralobular vessels
Positive GLUT1 and Lewis Y antigen	Negative GLUT1 and Lewis Y antigen	Negative GLUT1 and Lewis Y antigen

*Occasionally associated with thrombocytopenia during the neonatal period.

Vascular Malformations

- Vascular malformations are localized defects (errors) in vascular morphogenesis during embryonic development
- **Classification**
 - type of abnormal vessels (capillary, venous, lymphatic, arterial, arteriovenous, combined)
 - flow properties (slow-flow or fast flow)

Vascular malformation

Capillary malformation

1. PWS
2. Salmon patch
3. Hereditary hemorrhagic telangiectasia (HHT)
4. Cutis marmorata telangiectatica congenita (CMTC)

Venouse malformation

1. Blue rubber bleb
2. Maffucci syndrom

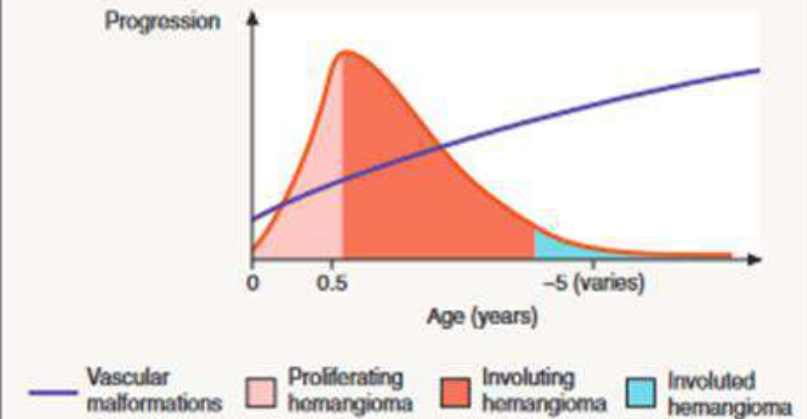
Lymphatic mal formation

1. Lymphangioma circumscriptum.
2. Cystic hygroma

DIFFERENCES BETWEEN VASCULAR MALFORMATIONS AND INFANTILE HEMANGIOMAS

Characteristics	Vascular malformations	Infantile hemangiomas
Clinical features	CM: red, macular during infancy VM: blue, compressible, fills with dependency LM: vesicles or large cysts AVM: four stages: stage 1 – dormant: red, warm; macular (mimics CM) or slightly infiltrated (mimics involuting infantile hemangioma) stage 2 – expansion: warm mass with thrill/throbbing, dilated draining veins stage 3 – destruction: pain, ulceration; occasional bone lysis and fracture stage 4 – cardiac decompensation; high-output cardiac failure; also signs of stage(s) 2 + 3	Premonitory markers at birth in 50%: red telangiectatic, bluish bruise-like, or white/anemic macules Superficial hemangioma: bright red, "strawberry"-like papules and plaques (during proliferative phase) Deep hemangioma: warm, rubbery mass under normal-colored or bluish skin
Sex prevalence	Female: male ratio 1:1	Female: male ratio 2:1 to 5:1
Natural history	Usually evident at birth Lifelong persistence, with commensurate growth during childhood and often slow worsening over time AVM: may worsen with puberty, trauma, and pregnancy	Precursor lesion or no evidence at birth Postnatal proliferation over 5–9 months, then slow involution over years Reach stable involuted stage
Radiology	VM or LM (slow flow): hypersignal on MRI T2-weighted sequence VM: thrombi and phleboliths (MRI, CT); gadolinium enhancement on MRI AVM (fast flow): flow voids on MRI T1- and T2-weighted sequences; arteriovenous shunts on Doppler ultrasound	Well-defined mass Some flow voids on MRI T1-weighted sequences (high-flow vessels)
Pathology (see Fig. 104.23)	CM: ectatic capillaries VM or LM: distorted, interconnected malformed channels AVM: AV fistulae and increased capillaries Typically no increase in cellular turnover	Dense lobular proliferation of endothelial cells forming capillaries with tiny lumens Increased cellular turnover in proliferating lesions can be demonstrated by elevated levels of markers of proliferation (PCNA, Ki-67), ECM remodeling (collagenase, urokinase), and vascular growth (bFGF, VEGF)
Immunophenotype	GLUT1-negative	GLUT1-positive
Hematology	VM or LVM: chronic LIC, occasional DIC Treatment of flares with low-molecular-weight heparin Possible benefit of low-dose aspirin as prophylaxis	(Kasabach-Merritt phenomenon is not linked to infantile hemangiomas, but to other vascular tumors such as kaposiform hemangioendotheliomas and tufted angiomas, both of which are GLUT1-negative; see Ch. 114)
Molecular biology	CMs: somatic <i>GNAQ</i> (including SWS) or less often <i>GNA11</i> mutation Sporadic VMs: somatic <i>TEK</i> (~50%) or <i>PIK3CA</i> (~25%) mutation LMs: somatic <i>PIK3CA</i> mutation Familial vascular malformations and overgrowth syndromes: see Tables 104.2 and 104.5	Rare families with autosomal dominant inheritance linked to 5q, and somatic loss of heterozygosity of 5q in some hemangiomas Germline mutation in <i>VEGFR2</i> or <i>TEM8</i> in some patients Somatic mutation in <i>VEGFR2</i> or <i>FLT4/VEGFR3</i> (5q34) in some hemangiomas Clonality in some hemangiomas

NATURAL HISTORY OF INFANTILE HEMANGIOMAS AND VASCULAR MALFORMATIONS

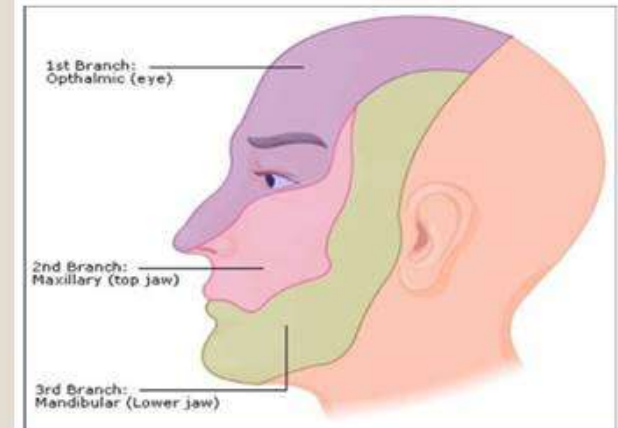


Port-wine stain nevus flamus

- PWS is a low-flow capillary malformation.
- at birth
- well-demarcated red macules and patches.
- The most common location is the face
- in the trigeminal nerve
 - V1: ophthalmic region
 - V2: maxillary region
 - V3: mandibular regio



Trigeminal nerve branches



- **H.p :**

- Widely dilated, thin-walled capillaries scattered throughout the upper dermis or the entire reticular dermis.

- **DD**

- Salmon patch
 - Precursor of infantile hemangiomas.

- **Course:**

- PWSs **do not involute spontaneously** and continue to grow over time



- **Complications**

-

1. Overgrowth of maxilla and other facial bones underlying a PWS .
2. ENLARGED GUM ,LIPS >>>>> gingival bleeding and lip incompetenc
3. lumbosacral area) may >>>>>> occult spinal dysraphism.
4. Syndromes associated with PWS



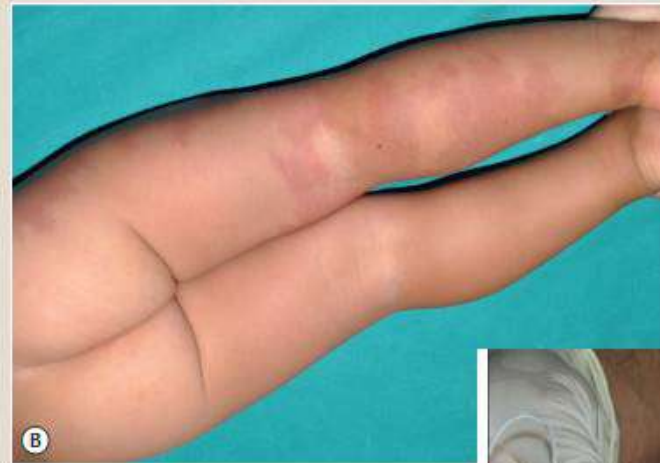
Sturge-Weber Syndrome (Encephalotrigeminal angiomatosis)

- **Triad of Capillary malformation (CM) of:**
 - Skin (PWS of at ophthalmic br. of trigeminal N.)
 - Ipsilateral leptomeningeal (Pia mater)CM.
 - Ipsilateral ocular CM
- **Phakomatosis**
- **Mainly v1 ,v2**



Klippel-Trenaunay syndrome (KTS):

- a combination of
- a limb CVM or
- CLVM
- with progressive overgrowth of the affected extremity
- lipomatosis, lymphedema,



PHAKOMATOSIS PIGMENTOVASCULARIS (PPV)

Type	Descriptive name	Proportion of reported cases	Major cutaneous components	Other reported cutaneous components
Ia/b	*	<5%*	Epidermal nevus + PWS	
IIa/b	Phakomatosis cesioflammea	75%	Dermal melanocytosis (aberrant Mongolian spots) + PWS	Nevus anemicus, hypotrichosis, lipohypoplasia, hypoplastic nails
IIIa/b	Phakomatosis spillosea or melanorosea	10%	Nevus spilus (macular type) + light red-pink CM ("nevus roseus")	Nevus anemicus, lymphedema, hypotrichosis
IVa/b	Phakomatosis melanovascularis or unclassifiable	10%	Café-au-lait macule + CM Variable, sometimes with features of II + III or II + V	Nevus anemicus, nevoid hyper- or hypopigmentation, nevus sebaceus
Va/b	Phakomatosis cesiomarmorata	<5%	Dermal melanocytosis + cutis marmorata telangiectatica congenita	

*Existence as a type of PPV has been challenged considering that epidermal nevi are not melanocytic lesions; epidermal nevi can occur together with vascular malformations in Proteus hamartoma tumor, and CLOVES syndromes (see Table 104.5), which are not generally considered as PPV.



Fig. 104.6
Phakomatosis pigmentovascularis type 2a (cesioflammea). A large port-wine stain and extensive dermal melanocytosis are seen.

OVERGROWTH SYNDROMES WITH VASCULAR MALFORMATIONS						
Gene	DCMO	PIK3CA-Related Overgrowth Spectrum			Proteus	PHTS
		M-CM	KTS	CLOVES		
	? <i>GNA11</i> (in some patients)	<i>PIK3CA</i> , mosaic	<i>PIK3CA</i> *, mosaic	<i>PIK3CA</i> , mosaic	<i>AKT1</i> , mosaic	<i>PTEN</i> germline with somatic second mutation
Asymmetric/segmental overgrowth						
At birth	+	+	+	+	–/minimal	+
Progressive	–	–	+	+	+++	+
Correlates with site(s) of vascular malformation	–	–	+	+	–	variable
Digital abnormalities	30%: syndactyly, sandal-gap, macrodactyly	Syndactyly > polydactyly, macrodactyly; sandal gap	Occasional macrodactyly, syndactyly (associated with deep venous anomalies)	Broad hands/feet, macrodactyly (especially 3 rd toe), sandal gap	Macrodactyly with distorted/deformed digits	–
Macrocephaly	–	+: (hemi) megalencephaly, dolichocephaly	–	30% with (hemi) megalencephaly	Dolichocephaly	+
Capillary malformations	Reticulated ± confluent areas; widespread, block-like	Reticulated; widespread, block-like; persistent nevus simplex	Geographic, well-demarcated, dark (associated with LM) or blotchy light; on extremity ± extension	Geographic, well-demarcated, dark; on trunk overlying lipomatous mass	Well demarcated; often on trunk	Variable
Venous anomalies	Variable prominent veins	Variable prominent veins	VM ^S ; often persistent lateral marginal vein, abnormal deep venous system	VM ^S of superficial, deep thoracic, and often major central veins	Variable VM ^S	Variable VM; dilated draining veins associated with high-flow lesions
Lymphatic anomalies	–	–	Often LM with overlying vesicles, lymphedema	Truncal LM within lipomatous mass; overlying vesicles	Variable LM	Variable LM, lymphedema



Fig. 104.10 Plantar cerebriform connective tissue nevus in a man with Proteus syndrome.

- **The 2 main lines of treatment**
- lasers (e.g. PDL)
- cosmetic camouflage.

(2) Salmon patch

- A common capillary malformation (present in approximately 40% to 70%
- Erythematous patches located on the mid-face
- , eyelids, and nape or occiput
- Prognosis: fades spontaneously (face) - may persist for





Fig. 104.12 Cutis marmorata telangiectatica congenita (CMTC). Note the hypoplasia of the affected limb (A, B) and associated cutaneous atrophy leading to a depressions (C). The color of the broad vascular network can be red-purple (A, C) or, as seen in this infant of Caribbean descent, brownish purple (B).
C, Cleary, Richard Aronow, MD



Hereditary hem orrhagic telangiectasia (HHT)

Osler-Weber-Rendu disease,

- mucocutaneous telangiectasias
 - visceral AVMs
- that have the liability to bleed.
- The first manifestation of HHT is usually **epistaxis**





Fig. 104.11 CLAPD syndrome. **A** Capillary malformation on the lower lip and chin, with swelling of the neck due to a lymphatic malformation. **B** Microcystic lymphatic malformation on the tongue. *A, Courtesy, Julie V Schaffer, MD.*

Venous malformations

- Blue hue, soft consistency, compressibility and tendency to fill with dependency
- Large VMs may be associated with:
 - Pulmonary thromboembolism
- DD
 - Deep infantile hemangiomas
 - KTs
 - Doppler ultrasound ,MRI
 - TREATMENT
 - compression
 - Aspirin.
 - Low molecular weight heparin .
 - Sclerotherapy followed by surgical resection.
 - Lasers: pulsed dye and/or Nd:YAG lasers



Fig. 104.13 Venous malformations (VMs). A Distortion of the tongue and lower lip has led to an open bite. B The skin and muscles of the entire arm are affected, with soft bluish nodules and obvious swelling in the dependent position. C This VM has a segmental distribution on the right trunk and involvement of the ipsilateral arm.

Syndroms associated with VMs

1. **Blue rubber bleb nevus syndrome**

- small dark blue papules and nodules (rubber blebs)
- Larger VMs
- Gastrointestinal lesions that bleed and result in iron deficiency anemia.
-



Fig. 104.15 Blue rubber bleb nevus syndrome. Multiple scattered small blue venous nodules, several of which are superimposed on a larger subcutaneous venous malformation. Courtesy, Juan Carlos López, MD.

Maffucci syndrome

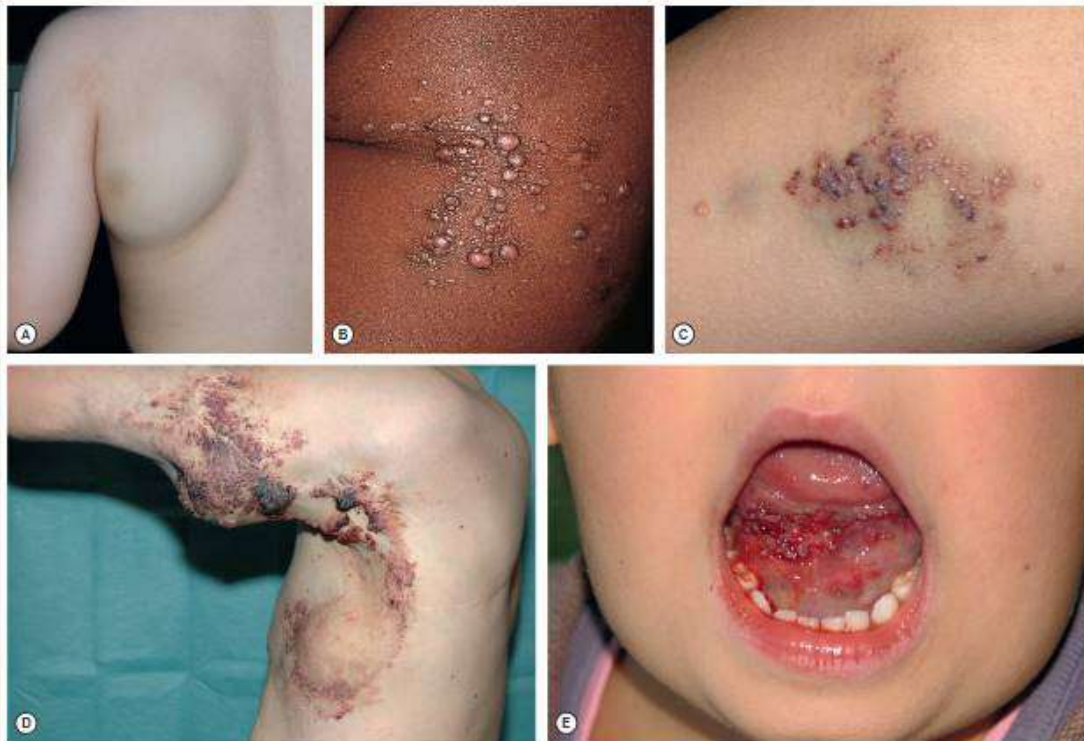
- VMs (blue to skin-colored nodules)
- Enchondromas (hamartomatous proliferations of chondrocytes)
- It most commonly affects the extremities



Fig. 104.16 Maffucci syndrome. **A** Both nodular venous malformations and enchondromas distort the hand. **B** A radiograph demonstrates an enchondroma and multiple phleboliths.

Lymphatic malformations

- Superficial lymphatic malformation
 - **Lymphangioma circumscriptum**
 - The most common type of LM.
 - Appears at birth or shortly after.
 - tense grouped translucent vesicles similar to frog's eggs (frog spawn) colourless fluid
 - axilla, shoulder, neck, **trunk** and tongue



○ ***Deep lymphatic malformation***

- cystic hygroma
- soft circumscribed subcutaneous swelling. Overlying skin is normal.
- neck, axilla, or groin



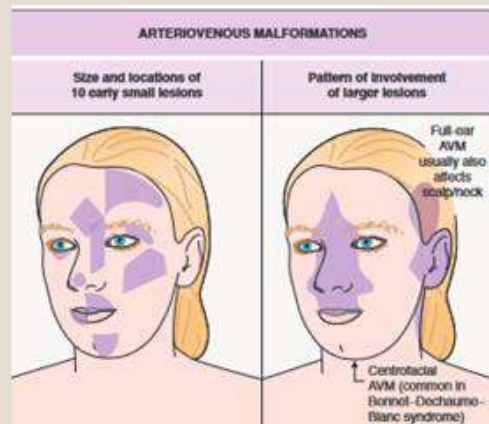


Fig. 104.20 Arteriovenous malformations - size, locations, and patterns.



Fig. 104.21 Cobb syndrome. This stain on the nape has an appearance reminiscent of a salmon patch, but it was warm and superimposed darker red macules progressively appeared. Doppler ultrasonography, MRI, and arteriography confirmed the presence of a metameric arteriovenous malformation of the skin and spine.



Fig. 104.19 Arteriovenous malformations (AVMs). A, B Dominant (stage 1) AVMs mimicking a port wine stain (A) and infantile hemangioma (B). C Expanding (stage 2) AVM on the external ear. D Destructive (stage 3) AVM that has led to cutaneous necrosis. (C. Goering, Juan Carlos Lopez, MD)

